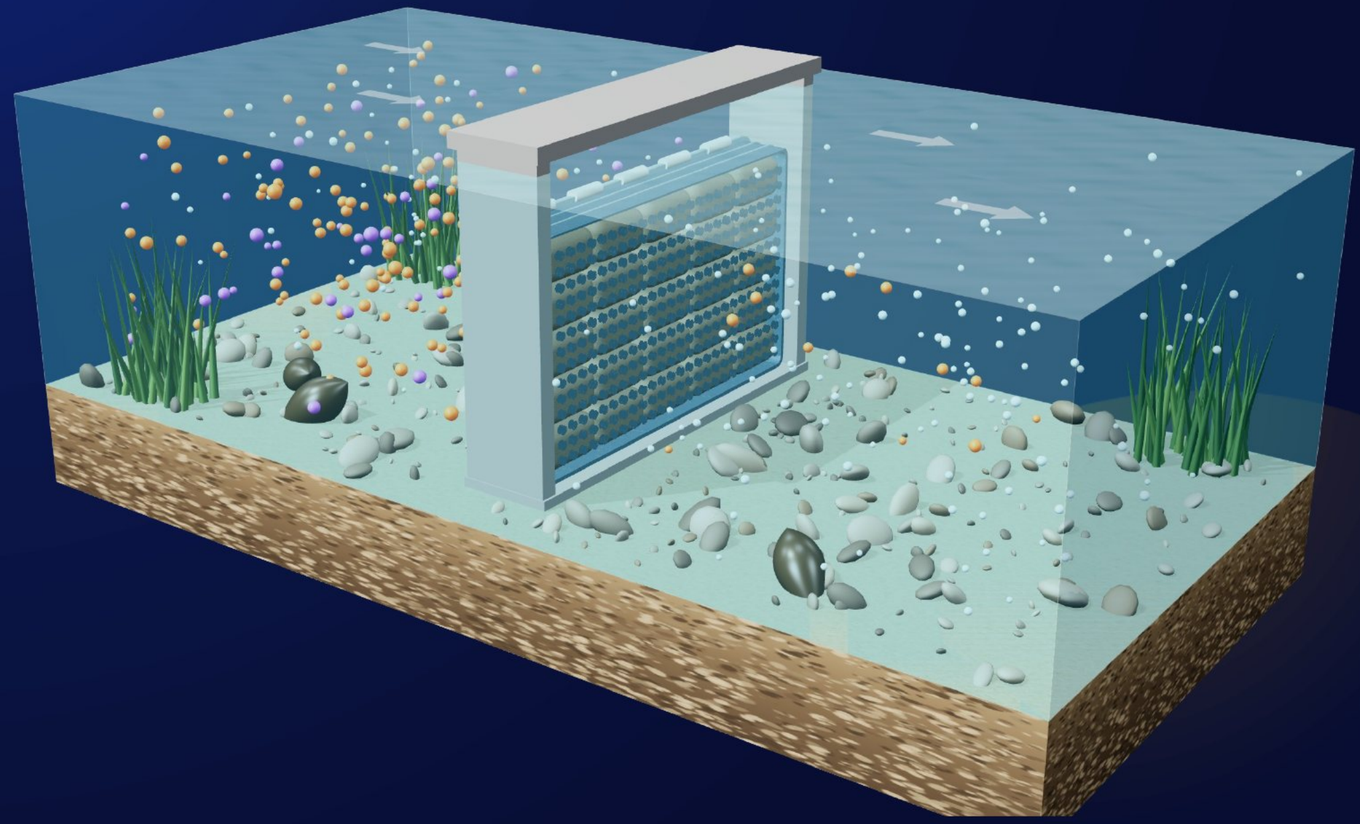
T.C. GENÇLİK VE SPOR BAKANLIĞI · TÜRKİYE CUMHURİYETİ ÇEVRE, ŞEHİRCİLİK VE İKLİM DEĞİŞİKLİĞİ BAKANLIĞI · COP31 TÜRKİYE ANTALYA · Climate High-Level Champions · SIFIR ATIK VAKFI · COP31 GENÇLİK İKLİM İDEATHONU · unicef her çocuk için

GROUP 9 · ERZURUM

BİYO-MİDYE

An electricity-free filter cassette that protects water at its source



THEME 5
Fighting Forest Fires, Drought and Water Stress

PROBLEM: WATER CRISIS

Türkiye’s water is both shrinking and getting polluted

USABLE WATER PER PERSON

1,308 m^3



We are below the water-stress line and get a little closer to water scarcity every year.

Drought

In dry years dams drop to critical levels; big cities are left with only a few months of water.

Erratic rainfall

Drought one year, floods the next: we cannot rely on how much water we will have.

Polluted water

The water we have left is getting polluted. Polluted water can be neither drunk nor used for irrigation.

Wildlife

Fish deaths in streams threaten biodiversity.

OUR MAIN GOAL

Protect water at its source

Drinking water · Agriculture · Wildlife · Economy

Sources: DSİ (Milliyet, 19.07.2025) · Falkenmark 1989 · MGM water-year data (2025, 2026) · Evrensel, 15.08.2025 · Habertürk, 26.12.2024

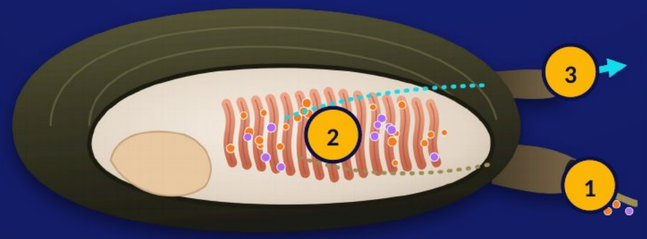
OUR INSPIRATION: THE MUSSEL

A mussel filters water and keeps pollution in its gills. So does Biyo-Midye.

IN NATURE: FRESHWATER MUSSEL



A real freshwater mussel



1. **Water enters the siphon**
2. **Gills trap particles**
3. **Filtered water leaves**

One freshwater mussel filters **up to half a litre** of water an hour.

OUR VERSION: THE BIYO-MIDYE CASSETTE



1. **Perforated lid + pumice pre-layer** = siphon
2. **Zeolite + biochar bags** = gills
3. **Cleaned water leaves** = filtered water

Source: Malacological Society of London, “Freshwater mussels as biofilters” (Aldridge, Univ. of Cambridge) · Cassette image: the team’s 3D model

RECIPE AND BUDGET

What is in one cassette, and what does the pilot cost?



4.8 kg

Natural zeolite

60% of bag volume · traps ammonium



1.4 kg

Biochar

40% of bag volume · traps metals



≈ 3 L

Pumice stone

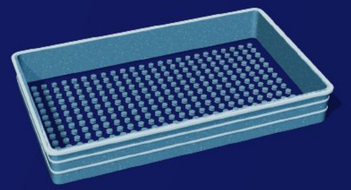
Pre-layer on the inlet face · catches sediment, also holds ammonium



≈ 1 m²

Geotextile felt

25 bags × 250 g filling



1 piece

Cassette body

Recycled HDPE · 50 × 30 × 9 cm

Item	Quantity	Unit price	Amount (TL)
Natural zeolite	25 kg	1,312 TL / 25 kg	**1,312**
Biochar	10 kg	830 TL / 5 kg	**1,660**
Pumice stone	20 L	≈ 190 TL / 20 L	**190**
Geotextile felt	5 m²	68.41 TL / m²	**342**
Cassette body (2) + steel frame	1 set	workshop quote (estimate)	**2,500**
Water analysis: ammonium, nitrite, nitrate	2 samples	2,469 TL / sample	**4,938**
PILOT TOTAL (first 90 days)			**≈ 10,940**

Filling for 1 cassette ≈ 580 TL

Pilot ≈ 10,940 TL: within the TÜBİTAK 2209-A limit of 12,000 TL.

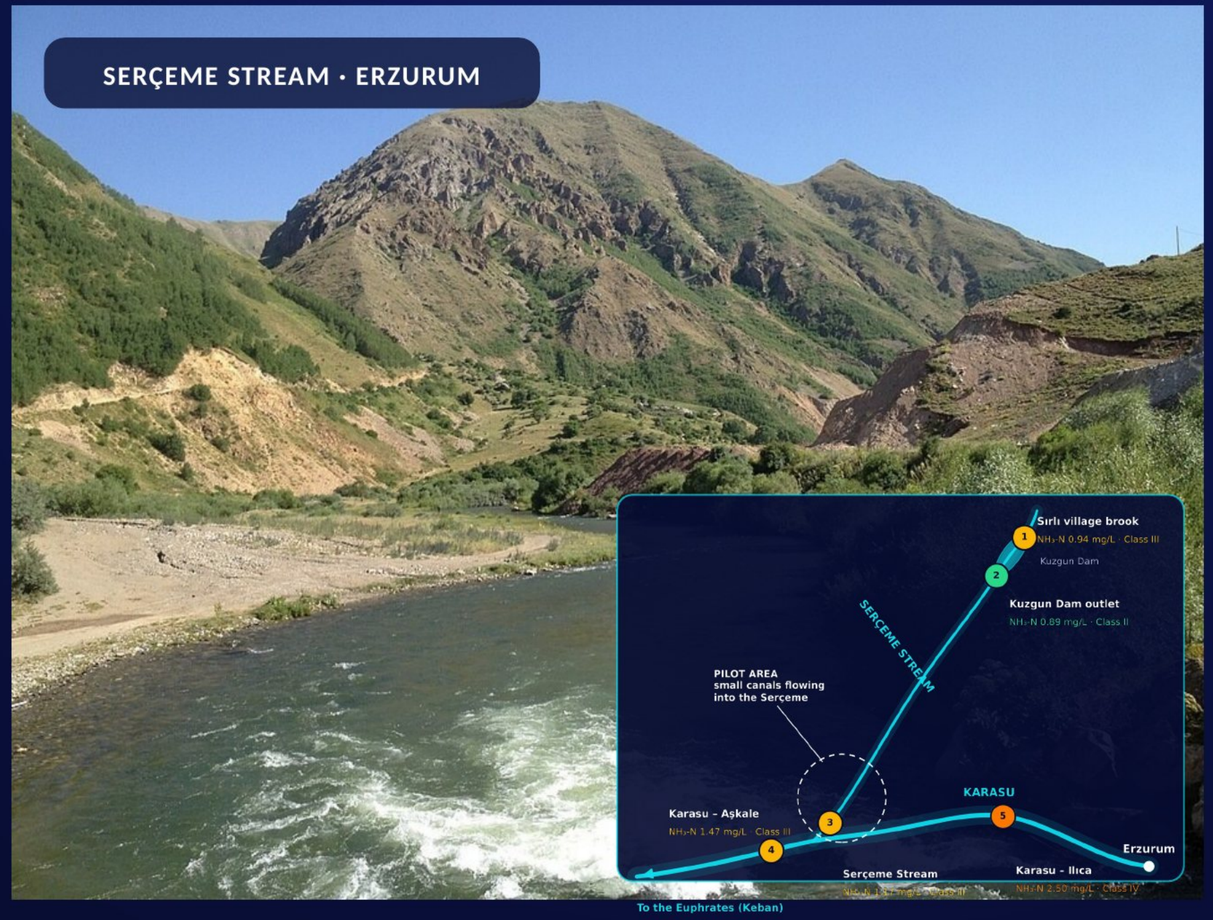
"Used filter material is an input for organic farming; this matters for the economic cycle."

Ahmet Faruk Can Özkan, Agricultural Engineer (AGRIVERTS)

Prices October 2026: Kimya Depom (zeolite 25 kg) · Biorfe (biochar 5 kg) · Sincan Peyzaj (pumice) · ÇŞB 2026 unit price 15.245.1002 (felt 250 g/m²) · Ministry of Agriculture and Forestry analysis list 2026. Body + frame: estimate, quotes to follow.

PILOT AREA

Serçeme Stream: a headwater of the Euphrates, where it joins the Karasu



1.17 mg/L

Ammonia nitrogen

annual mean · limit 1.0

4.5×

Nitrite nitrogen

4.5 times the limit

4 / 5

Sampling points

Class III; Karasu Class IV



BIODIVERSITY

Mass fish deaths in the Karasu at Aşkale: **January 2022, December 2024.**

Ammonia is toxic to fish and mussels.

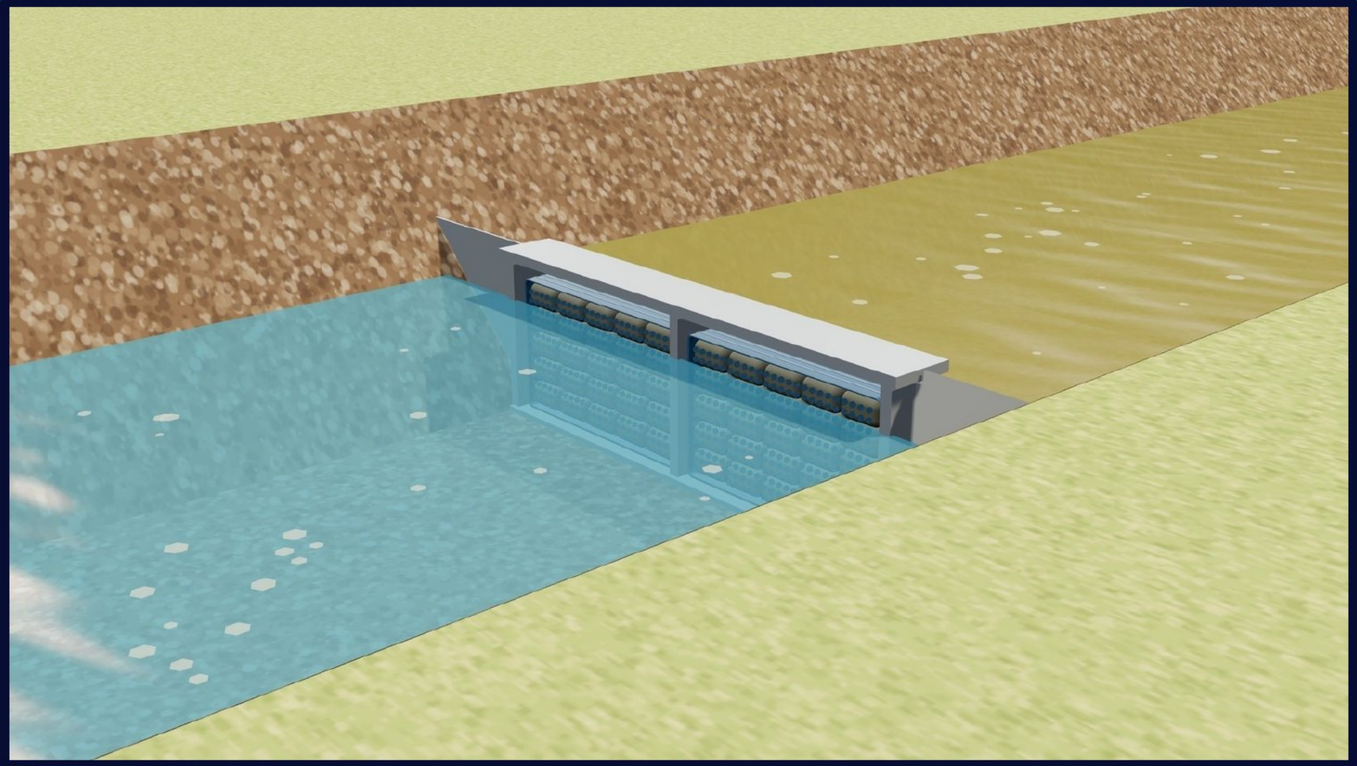
WHY HERE?

— 11 months of data from 5 sampling points (Atatürk Univ.)
— Pollution travels via the Karasu to the Euphrates and Keban Dam
— Natural nickel-chromium in the valley: we measure metals separately

Sources: Arslan 2021, Atatürk Univ. PhD thesis (June 2019–May 2020), Tables 5, 13–16 (schematic map) · Photo: Serçeme Stream · Surface Water Quality Regulation (OG 28483) · Bilgin 1984 · Fish deaths: İHA (17.01.2022), Habertürk (26.12.2024)

IN THE FIELD

How is it fitted to a canal, and how does it work under water?



TECHNICAL: 50 × 30 × 9 cm cassette · 25 bags × 250 g (60% zeolite + 40% biochar) · 3 L pumice pre-layer · recycled HDPE · steel frame

WASTE: full bags are collected in a closed container and analysed; if they contain metals they go to a **licensed hazardous-waste facility**.

1 Frame is driven in
Side plates guide water into the cassettes

2 Cassettes go in
Two cassettes; by hand, same day

3 Water flows through
Ammonium and metals stay in the bags

4 Bags are replaced
Full bags are sent for analysis

POSITIVE IMPACTS

+ Less ammonia and metal enters the stream
+ Fish, mussels and biodiversity are protected
+ Villagers join the monitoring; waste re-enters the cycle

No power · No pump · No chemicals

IMPACT

One cassette, with no electricity, cleans ammonia from 2,300–4,300 litres of water

2,300–4,300 litres

one cassette, until its bags are replaced · 2.7–5.0 g of nitrogen kept out of the stream

= **9–17 days** of one person's tap water (1 drop = one person's daily water, 255 L)

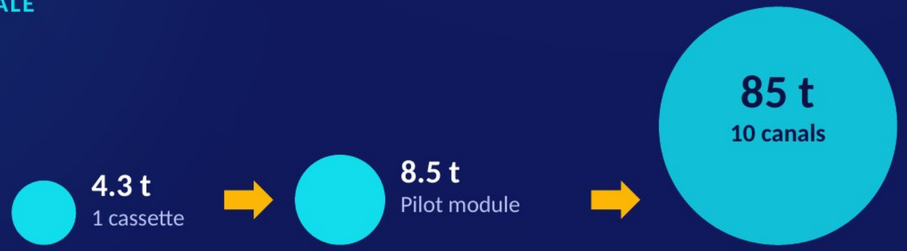


190–360

mussels' worth of filtering in one day

The water one cassette cleans equals a day's filtering by 190–360 mussels. (1 mussel ≈ 0.5 L/hour)

AT SCALE



WHAT MAKES US DIFFERENT

- **0 kWh** no power, pump or chemicals
- **Same day** carried and installed by hand
- **Waste in the loop** analysis → licensed facility / compost

Environmental

Less ammonia and metal reaches the stream; fish, mussels and biodiversity are protected. No electricity.

Social

Villagers and students take part in monitoring; water awareness grows in the village.

Economic

Cassette filling ≈ 580 TL; no big treatment-plant investment. If the analysis is clean, the filling becomes compost.

Calculation: 4.8 kg zeolite × 0.55–1.03 mg N/g ÷ 1.17 mg/L (Boulinguiez 2005; Arslan 2021) · TÜİK 2024: 255 L/person·day · Mussel ≈ 0.5 L/hour (Malacological Society); the mussel comparison is an analogy.

IMPLEMENTATION

With whom, in which steps, against which risks?

STAKEHOLDERS

TÜBİTAK	First funding via 2209-A (up to 12,000 TL)
SIFIR ATIK VAKFI	Waste cycle and scaling up
TÜRKİYE CUMHURİYETİ ÇEVRE, ŞEHİRCİLİK VE İKLİM DEĞİŞİKLİĞİ BAKANLIĞI	Water quality monitoring and regulations

ROADMAP

DAYS 1–30
Measure and choose

DAYS 31–60
Laboratory test

DAYS 61–90
Prototype

APRIL–OCTOBER 2027
Field pilot

RISK → MEASURE

Risk	Measure	Source
Zeolite may hold less in hard water	Column test with Serçeme water	Boulinguiez 2005 · Serçeme hardness 128 mg/L (Arslan 2021)
The stream freezes in February	Seasonal operation April–October	Arslan 2021: February sampling was impossible because of ice
Natural nickel-chromium can skew results	Measure the natural background first	Bilgin 1984: Ni mean 2,240, Cr mean 1,812 ppm
Asbestos (chrysotile) in valley rocks	Field team does not break rocks, wears masks	Bilgin 1984: chrysotile in serpentinites
Bags may clog with sediment	Pumice pre-layer, weekly checks	design measure · Moradi et al. 2016
Used filling may contain metals	Analysis; licensed hazardous-waste facility	Waste Management Regulation (OG 02.04.2015, 29314)

ÇŞİDB: Ministry of Environment, Urbanisation and Climate Change. Scaling: Serçeme → other canals flowing into the Karasu → Euphrates basin.



TEAM

Group 9

COP31 Youth Climate Ideathon · Erzurum

- **F** Furkan
- **İ** İlayda
- **S** Sude
- **Z** Zeynep
- **H** Hasan

Teamwork: the problem, field interview, data analysis, design and calculation decisions are all our team's own.

AI: we used it for research and visuals; we checked the sources ourselves.

GROUP 9 · BİYO-MİDYE

Saving water at source.

An electricity-free, replaceable, measurable answer to water scarcity: a Biyo-Midye for every polluted canal.

WHAT WE NEED

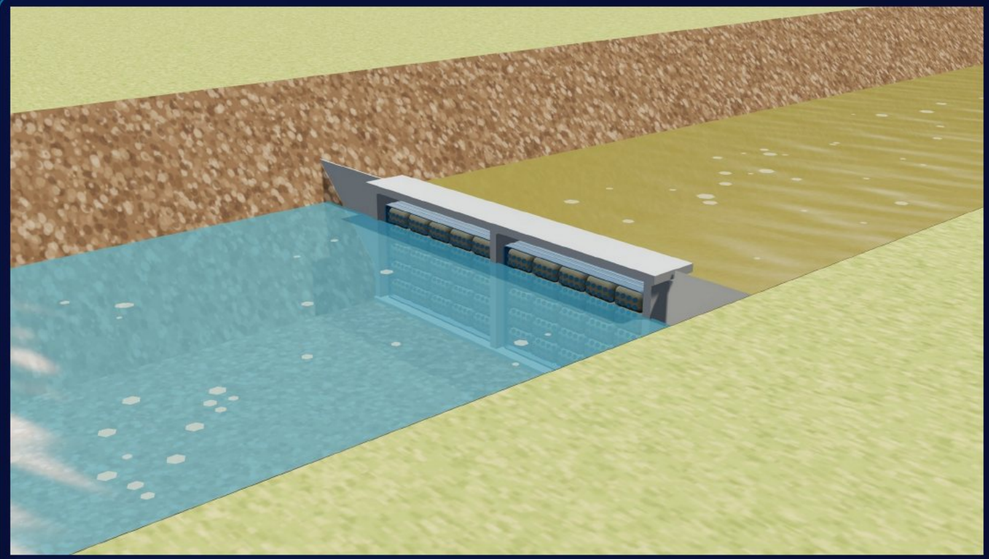
A university laboratory and an advisor for a column test with Serçeme water.

OUR FUTURE GOALS

If results are strong, prototypes in other polluted basins such as the Büyük Menderes.

Full filters are collected under control like medical waste and handed to a licensed hazardous-waste facility.

If the analysis is clean, the filling becomes organic compost material for organic farming.
(suggested by Agr. Eng. A. F. C. Özkan)



Pilot module installed in a canal (3D model): cleaned water in front

INTERACTIVE SIMULATION

Scan the QR code: change the flow, pollution and outlet weir yourself and watch the readings live.

furkanselimozkan07.github.io/My_AI_Projects/biyo-midye

REFERENCES

Problem and field data

1. Arslan, H. Y. (2021). Epiphytic and epilithic diatom communities of the Serçeme Stream (Karasu River)... PhD thesis, Atatürk University. YÖK Thesis No 676271.
2. Bilgin, A. (1984). Mineralogy and petrography of the Serçeme (Erzurum) Stream ultramafics. Jeoloji Mühendisliği, 81–87.
3. Nuroğlu, Ş., Kocaman, E. M. (2024). Effect of the Erzurum wastewater treatment plant on Karasu water quality. J. Agricultural Production, 5(3), 185–191.
4. US EPA (2013). Aquatic life ambient water quality criteria for ammonia – freshwater.
5. DSİ data: usable water per person 1,308 m^3 (2024). Milliyet, 19.07.2025. DSİ (2023) water use.
6. Falkenmark, M. (1989). The massive water scarcity now threatening Africa. Ambio, 18(2), 112–118.
7. Personal interview: A. F. C. Özkan, Agricultural Engineer, AGRIVERTS (October 2026).
8. Fish deaths in the Karasu at Aşkale: İHA / Son Dakika, 17.01.2022; Habertürk, 26.12.2024.
9. MGM water-year rainfall data: Bloomberg HT, 12.08.2025 and 11.06.2026. Dam levels: Evrensel, 15.08.2025.
10. TÜİK (2025). Water and Wastewater Statistics 2024.

Technology, inspiration and cost

11. Malacological Society of London. Freshwater mussels as biofilters (Aldridge, Univ. of Cambridge).
12. Boulinguiez, B. (2005). Ammonium removal in drinking water using natural zeolite. Kiwa Water Research, BTO 2005.052.
13. Wang, X. et al. (2020). Recent advances in biochar application for water and wastewater treatment. PeerJ, 8, e9164.
14. Ministry of Agriculture and Forestry, Soil, Fertilizer and Water Resources Central Research Institute (2026). Analysis price list.
15. Vendor prices, October 2026: Kimya Depom (zeolite 25 kg); Biorfe (biochar 5 kg); Sincan Peyzaj (pumice).
16. ÇŞB (2026). Construction unit prices, item 15.245.1002 (250 g/m^2 geotextile felt).
17. Moradi, M. et al. (2016). Ammonium removal by pumice as a natural low-cost adsorbent. Archives of Environmental Protection, 42(2), 33–43.

Policy and support

18. Surface Water Quality Regulation. Official Gazette, 30.11.2012, No 28483.
19. National Water Plan 2026–2035. Presidential Decision 11063, OG 14.03.2026, No 33196.
20. Climate Law, Law No 7552. Official Gazette, 09.07.2025.
21. Zero Waste Regulation. Official Gazette, 12.07.2019, No 30829.
22. Waste Management Regulation. Official Gazette, 02.04.2015, No 29314.
23. TÜBİTAK (2026). 2209-A University Students Research Projects Support Programme.